

MUHAMMAD HARITS KHAIRIANSYAH

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D4 Automation Engineering student at Institut Teknologi Sepuluh Nopember with experience in Industrial IoT, Automation Systems, ESP32-based Monitoring, Node-RED, and Electrical Maintenance. Experienced in developing monitoring dashboards using Grafana and implementing industrial communication systems through MQTT and Modbus RTU protocols. Passionate about Industrial Automation, IoT Systems, and Electrical Control Engineering.

WORK EXPERIENCE

PT. Sumber Graha Sejahtera(Sampoerna Kayoe) - Batang, Indonesia
September 2026 – February 2027

- Developing a Computerized Maintenance Management System (CMMS) to digitize emergency work order (EWO) reporting, approval, and maintenance tracking.
- Developing web applications using React, Python FastAPI, and PostgreSQL for equipment management, maintenance history, and user access control.
- Designing a maintenance KPI dashboard to monitor MTTR, MTBF, downtime, and EWO status.
- Designing an MCDM-based approach to prioritize emergency work orders based on equipment criticality and operational conditions.

PT. Indoprima Gemilang - Surabaya, Indonesia
Jan 2026 - Jul 2026

- Developed Industrial IoT systems using ESP32, MQTT, Node-RED, and database integration.
- Designed and implemented energy monitoring systems using PZEM sensors for three-phase electrical systems.
- Performed maintenance and troubleshooting on industrial IoT devices and smart weighing systems.
- Assisted PCB design and hardware integration for monitoring applications.
- Conducted installation and commissioning of IoT-based monitoring systems on production equipment.

PROJECT

Plastic Waste Shredder Automation and Monitoring System
Project Manager | Institut Teknologi Sepuluh Nopember

- Developed an automated plastic bottle shredder equipped with a real-time monitoring and protection system.
- Designed and implemented an ESP32-based monitoring architecture integrated with Modbus RTU communication and MySQL database storage.
- Integrated multiple sensors for current, temperature, weight, and operational monitoring to improve machine reliability and safety.
- Developed Grafana dashboard for real-time visualization of machine performance, electrical parameters, and operational status.
- Implemented automatic motor protection and anti-jamming mechanisms to ensure continuous operation during shredding processes. Developed ESP32-based devices for packing data acquisition and QR Code-based lot identification.

IoT-Based Packing Monitoring System
Industrial Project | PT Indoprima Gemilang

- Integrated MQTT, Node-RED, and SQL Server to enable real-time data transmission, processing, and storage.
- Designed operator interfaces using LCD displays and push-button inputs for production data entry.
- Implemented Over-The-Air (OTA) firmware updates to simplify maintenance and reduce deployment downtime.
- Supported production monitoring digitalization through Industrial IoT implementation

Three-Phase Energy Monitoring System
Industrial Project | PT Indoprima Gemilang

- Developed electrical energy monitoring systems using ESP32 and PZEM sensors for three-phase power measurement.
- Implemented real-time data acquisition and monitoring architecture using Node-RED and database integration.
- Performed testing, calibration, and validation of electrical parameters to ensure monitoring accuracy.
- Assisted deployment and troubleshooting of monitoring devices within industrial environments.

IoT Access Security System & Automated Reporting
Industrial Project | PT Indoprima Gemilang

- Designing a database-driven digital verification access control system

- Integrating hardware, MQTT, Node-RED, and a database into a single real-time workflow.
- Implementing a non-blocking finite state machine in the firmware to control the key sequence
- Implementing automatic WiFi/MQTT reconnection to ensure system reliability in an industrial environment.
- Deployed directly in the field (realworld deployment) during the internship

EDUCATION LEVEL

Institut Teknologi Sepuluh Nopember - Surabaya
Bachelor Applied Degree in Automation Engineering 2023 – Present
GPA: 3.26/4.00

ORGANISATIONAL & VOLUNTEER EXPERIENCE

BEM Fakultas Vokasi ITS
External Relations Staff | 2024–2025

- Establish and maintain communication with external partners to support the implementation of the organization's work programs.
- Manage sponsorship and collaboration processes with various parties to support student activities.
- Collaborate with the team to develop strategies for external relations and organizational networking.
- Coordinated the implementation of a TOEFL preparation program, including planning, team management, and event evaluation to ensure effective execution.

HMDTEO ITS
Himpunan Mahasiswa Teknik Elektro Otomasi ITS
Staff PSDM | 2024–2025

- Design and implement capacity-building programs for members through training and mentoring activities.
- Participate in the leadership development and competency-building processes for Electrical and Automation Engineering students.
- Coordinate with various divisions to support the effective implementation of organizational activities.

UKM Tenis Meja ITS
Head of External Relation Departemen

- Led external partnership and collaboration initiatives to support organizational activities and development programs.
- Coordinated division members in executing external communication strategies and partnership opportunities.
- Managed stakeholder relations to expand organizational networks and event support opportunities.

VOLUNTEER EXPERIENCE

Gorontalo Baik Indonesia (GOROBA)
Assessor Division | 2025–Present

- Participated in community service initiatives and assessment activities while collaborating with volunteers from diverse backgrounds to support social impact programs.

SKILLS

- **Technical skills:** Arduino IDE, SketchUp, GX Works, and Eagle (Electrical Wiring), MsSql, Data Monitoring and Analysis, Proteus, Learning 3D Modeling Fusion 360, Coppelia Sim EDU
- **Soft skills:** Problem Solving, Time Management, Orientation to Detail, Adaptability, Analytical Thinking and Teamwork
- **Languages:** Indonesian (Native), English (Basic Working Proficiency)

ACHIEVEMENTS & OTHER EXPERIENCE

- Junior Mobile Programmer Certification-Vocational School Graduate Academy, Digital Talent Scholarship – BPSDMP Surabaya (2025)
- National Participant – Olimpiade Vokasi Indonesia (OLIVIA) IX (2025)
- Participant – Kongres Ikatan Keluarga Mahasiswa Vokasi Indonesia (IKMVI) (2025)
- PLC and Microcontroller Training – Department Teknik Elektro Otomasi (2024)
- Responsive Web Design Certification – freeCodeCamp